SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA4304

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: **Total Marks: 10**

Code of Conduct:

I _____P.Ranjithkumar (192411119)_____ (Name / Reg No) certify that this submission is my originalwork and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Ranjithkumar

WEB DEVELOPMENT STANDARDS IMPLEMENTATION

Scenario: A start-up is developing a modern web application. During development, the team encounters issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

1. Correct HTML Tag for Video Embedding

Answer: Option B — *<video src="video.mp4" controls></video>*

The HTML <video> element is used to embed video content in a webpage. The controls attribute displays playback controls such as play, pause, and volume. The other options (<media>, <movie>, and <vid>) are not standard HTML video elements.

Example: <video src="video.mp4" controls></video>

This allows visitors to play the promotional video directly on the homepage without requiring a separate video player application.

2. Appropriate HTTP Method for Form Submission

Answer: Option B — POST

POST sends submitted form values in the request body and is suitable for sending registration information to the server. GET places form values in the URL, which is not appropriate for sensitive registration details. POST alone does not encrypt data; HTTPS should be used to protect information during transmission.

Example: <form action="/register" method="post"> ... </form>

The server receives the request, validates the submitted values, and can then process the registration.

3. Proper Semantic Element for Navigation

Answer: Option B — <nav>

The <nav> element represents a section of a webpage that contains major navigation links, such as Home, About, Services, and Contact. It gives the page a meaningful structure and helps browsers and assistive technologies identify the navigation area.

Example:

```
<nav>\n  <a href="/">Home</a>\n  <a href="/about">About</a>\n  <a
href="/contact">Contact</a>\n</nav>
```

4. Correct XML Syntax and Errors in Other Options

Answer: Option B — *<note><to>User</to><from>Admin</from></note>*

XML requires elements to be properly nested, and every opening tag must have a matching closing tag. Option B has matching tags and a complete root element.

- **Option A:** The closing </note> tag is missing, so the root element is not closed.
- **Option C:** The <to> element is incorrectly closed with </from>, creating mismatched tags.
- **Option D:** The final closing tag is incomplete because the closing angle bracket (>) is missing.

5. Correct Protocol for Web Communication

Answer: Option B — HTTP

HTTP stands for Hypertext Transfer Protocol. It enables communication between a web browser and a web server. The browser sends requests for web resources, and the server returns responses containing status codes, headers, and content. HTTPS is the secure version of HTTP and is recommended when transmitting registration or other sensitive information.

Other options: FTP is mainly used for file transfer; SMTP is used for sending email; and SNMP is used for network monitoring and management.

Overall Conclusion

Following web development standards helps create reliable, accessible, and secure web applications. The <video> element embeds video content, POST submits form data in the request body, <nav> gives navigation links semantic meaning, valid XML uses properly matched tags, and HTTP supports communication between browsers and servers. HTTPS should be used wherever sensitive information needs protection during transmission.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25 (21)	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25 (20)	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20 (16)	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20 (16)	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10 (7)	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand